



LARANA, INC.

CS 555

MULTIPLE LINEAR REGRESSION PROJECT

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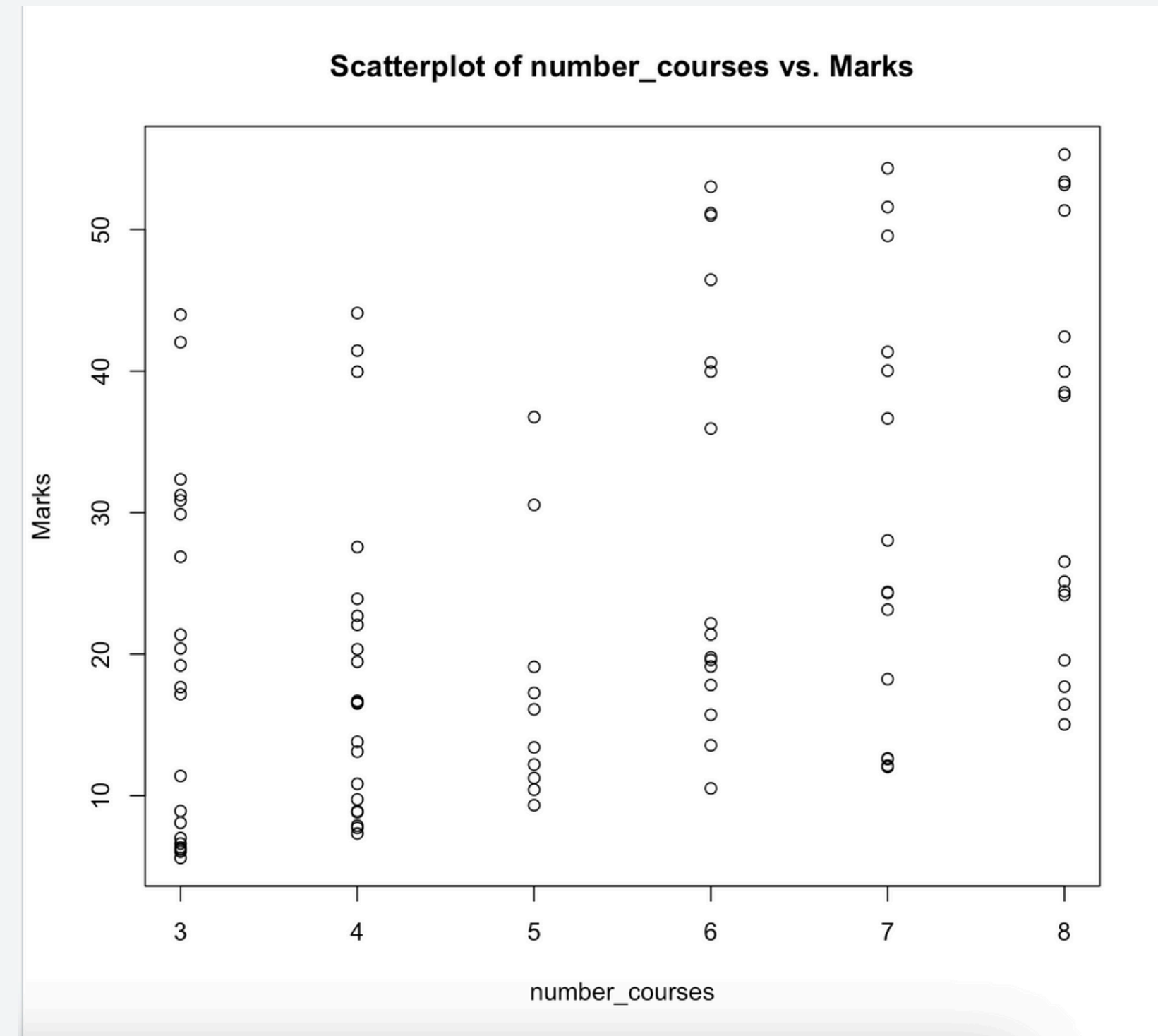
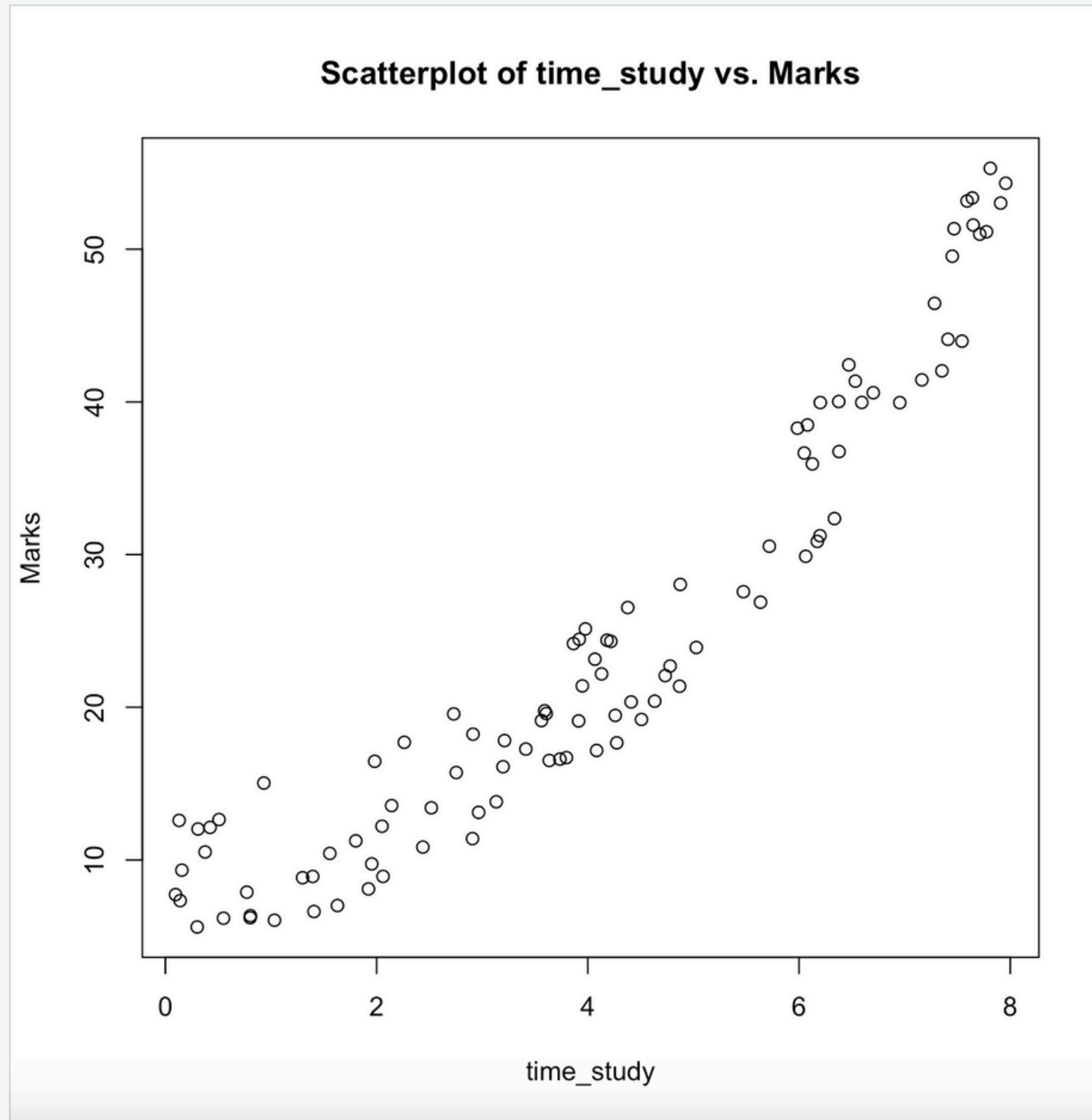
Research Scenario Description

The longer time students spending on study, the better students achieve good Marks in school. In order to study this phenomenon in more detail and to see if time_study associated with increase in school Marks. 100 students taking between 3 to 8 courses were randomly selected for time_study and Marks in school.

Data Resources:

<https://www.kaggle.com/datasets/yasserh/student-marks-dataset>

Association Between Marks and Time_Study and Number_Courses



The Strength of the graph is strong associated, which implies that the Marks will increase while time_study or number_courses increase. We can conclude this is a perfect one-to-one correspondence.

Methodology - Multiple Linear Regression

Functionality of MLR: Describe the relationship between a set of independent(explanatory) variables and a dependent(response) variable.

Independent(or explanatory) variables in this experiment:

time_study, number_courses

Dependent(or response) variable in this experiment:

Marks

MLR equation: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + e$

The Least-Squares Regression Equation

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-7.4563	1.1745	-6.349	6.98e-09	***
time_study	5.3992	0.1529	35.303	< 2e-16	***
number_courses	1.8641	0.2017	9.243	5.78e-15	***

The least-squares regression equation:

$$\hat{Y} = -7.4563 + 5.3992 * \text{time_study} + 1.8641 * \text{number_courses}$$

Assessing the Fit of the Regression Line

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{\text{Reg SS}}{\text{Total SS}}$$

The coefficient of determination concludes the percentage of the variation in the response variable explained by the multiple regression model.

$$R^2 = 0.9404$$

Conclusion: 94% of the variability in students' Marks can be explained by time_study and number_courses.

Set up Hypothesis

Null Hypothesis $H_0: \beta_{\text{time_study}} = \beta_{\text{number_courses}} = 0$

(time_study, number_courses are not significant predictors of Marks)

Alternative Hypothesis $H_1: \beta_{\text{time_study}} \neq 0$

or $\beta_{\text{number_courses}} \neq 0$

(at least one of the slope coefficients is different than 0; time_study or number_courses are significant predictor of marks. $\alpha = 0.06$)

Inference - F Test (Global Test)

	SS (Sum of Squares)	df (degrees of freedom)	MS (Mean Square)	F -statistic	p-value
Regression	Reg SS	Reg df = k	Reg MS = Reg SS / Reg df	$F = \text{Reg MS} / \text{Res MS}$	$P(F_{\text{Reg df}, \text{Res df}, \alpha} > F)$
Residual	Res SS	Res df = $n - k - 1$	Res MS = Res SS / Res df		
Total	Total SS = Reg SS + Res SS				

ANOVA table tests the alternative hypothesis that at least one of the slope parameters is different than 0.

$$F = \text{Reg MS} / \text{Res MS}$$

Decision Rule

The decision rule for a level α at 0.04:

- Reject $H_0: H_0: \beta_1 = \beta_2 = \dots = \beta_k = 0$ if $F \geq 2.896613$
- Otherwise, do not reject H_0
- Where $F(k, n-k-1, \alpha)$ is the value from the F-distribution table with k degrees of freedom(numerator), $n-k-1$ degrees of freedom(denominator) and associated with a right hand tail probability of α
 $F(2, 100-2-1, 0.06) = 2.896613$

F Statistic Test Result & P - Value

$$F = \text{Reg MS} / \text{Res MS} = 764.8$$

$$P\text{-value} = 0 < 0.06$$

R code:

```
#multiple_regression_model
```

```
multiple_regression_model <- lm(Marks ~ time_study + number_courses , data = data)
```

```
summary(multiple_regression_model)
```

```
#Critical value of F test
```

```
qf(1-0.06, df1 = 2, df2 = 97)
```

```
# p value
```

```
1-pf(764.8,df1=2, df2=97)
```

Conclusion

Reject H_0 since $F \text{ statistic} \geq 2.896613$. We have significant evidence at the $\alpha=0.06$ level that $\beta_{\text{time_study}} \neq 0$ and/or $\beta_{\text{number_courses}} \neq 0$. That is, there is evidence of a linear association between Marks and time_study and or number_courses.

(here, $p < 0.06$ as calculated using software program).

Inference - T Test

The overall model is significant. The next step is to perform testing on each individual parameter to identify the relative contribution of each independent variable.

$$t = \frac{\hat{\beta}_i}{SE_{\hat{\beta}_i}}$$

Use this T statistic model, controlling for the other independent variable in the model, we can test each if $\beta_i = 0$

Set up Hypothesis

Null hypothesis H_0 : $\beta_{\text{time_study}} = 0$ (after controlling for number_courses)

Alternative hypothesis H_1 : $\beta_{\text{time_study}} \neq 0$ (after controlling for number_courses)

$\alpha = 0.06$

Select the appropriate test statistic

$$t = \frac{\hat{\beta}_i}{SE_{\hat{\beta}_i}}$$

with $df = n - 3 = 100 - 3 = 97$ degrees of freedom

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-7.4563	1.1745	-6.349	6.98e-09	***
time_study	5.3992	0.1529	35.303	< 2e-16	***
number_courses	1.8641	0.2017	9.243	5.78e-15	***

The decision rule for a level α at 0.06:

- Reject $H_0: H_0: \beta_{\text{time_study}}=0$ if $|t| \geq 1.903045$
- Otherwise, do not reject H_0
- Where $t(n-k-1, \alpha/2)$ is the value from the t-distribution table with $n-k-1$ degrees of freedom and associated with a right hand tail probability of $\alpha/2$

$$t\text{-value}(97, 0.06/2) = 1.903045$$

$$> qt(0.97, df=97)$$

$$t(\text{time_study}) \text{ value} = 35.303$$

$$P\text{-value} = 2.2e-16 < 0.06$$

Conclusion - is time_study a significant predictor of Marks after controlling number_courses?

- Reject H0 since t statistic ≥ 1.903045 . We have significant evidence at $\alpha = 0.06$ level that $\beta_{\text{time_study}} \neq 0$. That is, there is evidence of linear association between Marks and time_study.
- Student Marks will increase 5.3992 for a one unit increase in the independent variable(time_study) after controlling for the other independent variables(number_courses).
- We can say with 94% confidence that the true value of β (time_study) is between(5.108224, 5.690176) after controlling for other independent variable number_courses.

$$\hat{\beta}_i \pm t_{n-k-1, \frac{\alpha}{2}} \cdot SE_{\hat{\beta}_i}$$

Set up Hypothesis

Null hypothesis H_0 : $\beta_{\text{number_courses}} = 0$ (after controlling for time_study)

Alternative hypothesis H_1 : $\beta_{\text{number_courses}} \neq 0$ (after controlling for time_study)

$\alpha = 0.06$

Select the appropriate test statistic

$$t = \frac{\hat{\beta}_i}{SE_{\hat{\beta}_i}}$$

- Decision Rule: Reject $H_0: \beta_{\text{number_courses}} = 0$ if $|t| \geq 1.903045$
- Otherwise, do not reject H_0

$t(\text{number_courses})$ value = 9.243

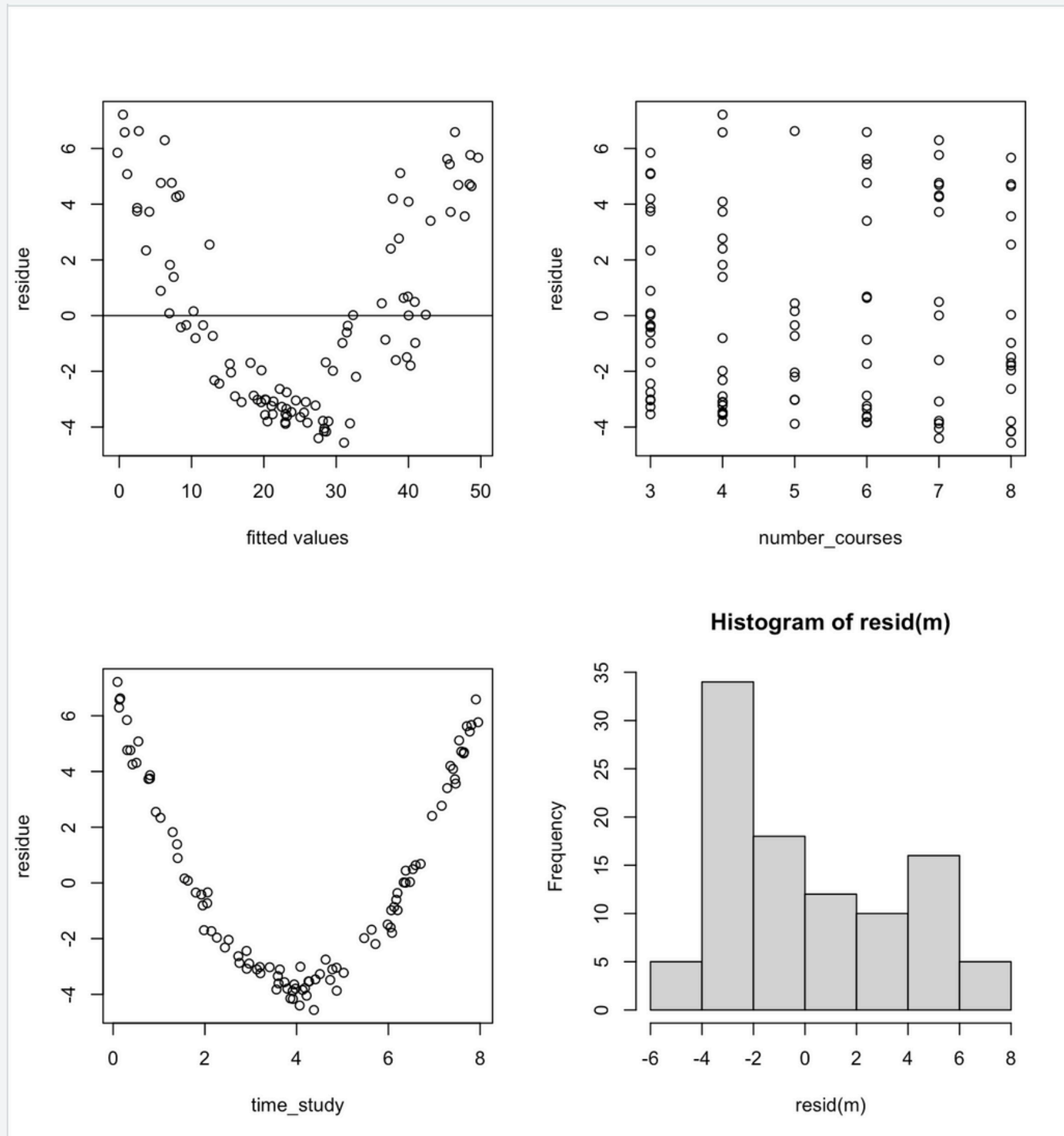
P-value = $5.78e-15 < 0.06$

Conclusion - is number_courses a significant predictor of Marks after controlling time_study?

- Reject H0 since t statistic ≥ 1.903045 . We have significant evidence at $\alpha = 0.06$ level that $\beta_{\text{number_courses}} \neq 0$. That is, there is evidence of linear association between Marks and number_courses.
- Student Marks will increase 1.8641 for a one unit increase in the independent variable(number_courses) after controlling for the other independent variables(time_study).
- We can say with 94% confidence that the true value of $\beta_{\text{number_courses}}$ is between(1.480256, 2.247944) after controlling for other independent variable time_study.

$$\hat{\beta}_i \pm t_{n-k-1, \frac{\alpha}{2}} \cdot SE_{\hat{\beta}_i}$$

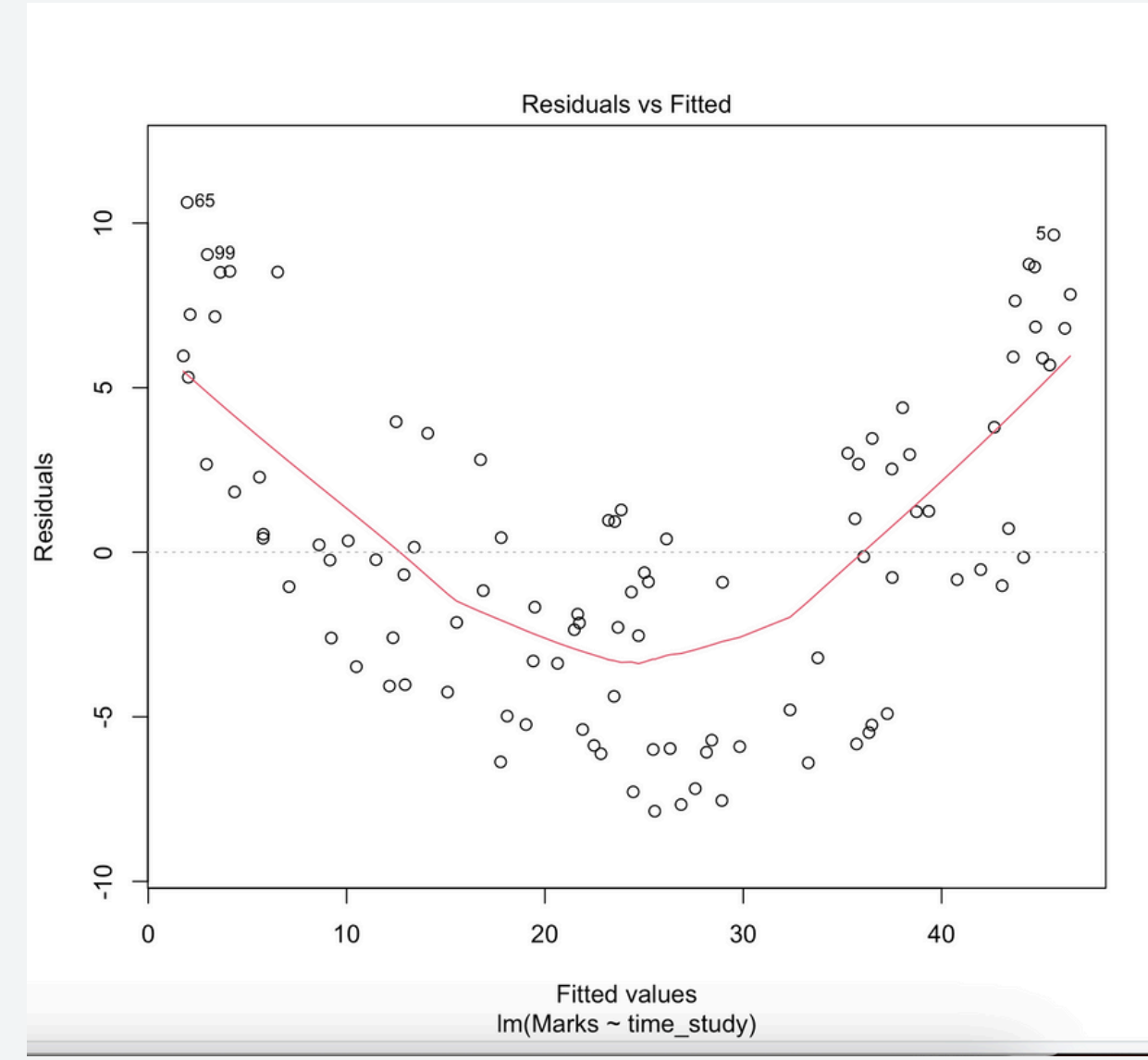
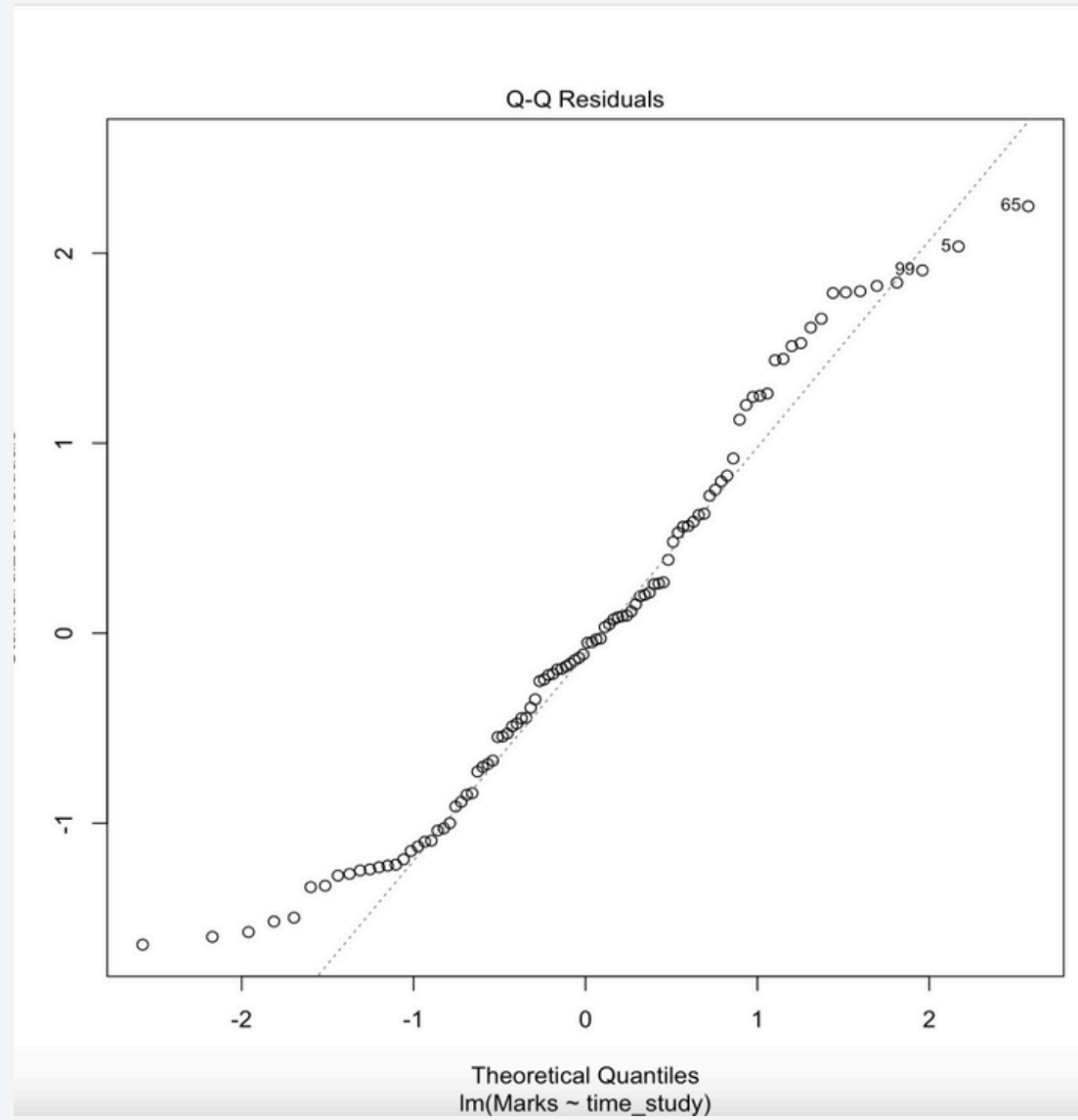
Limitations - Assess Whether the Model Assumptions are Met ?



Key Requirements:

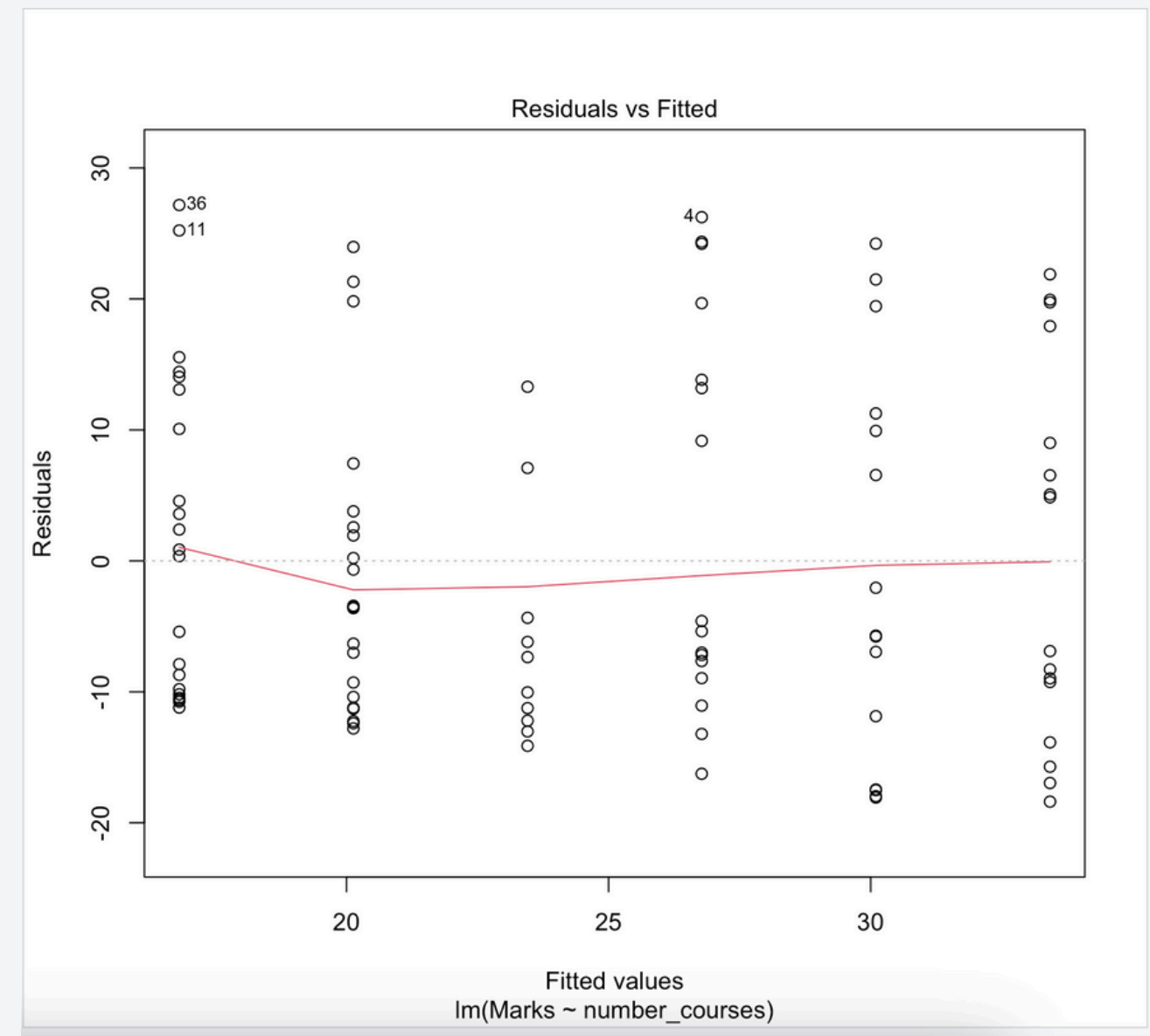
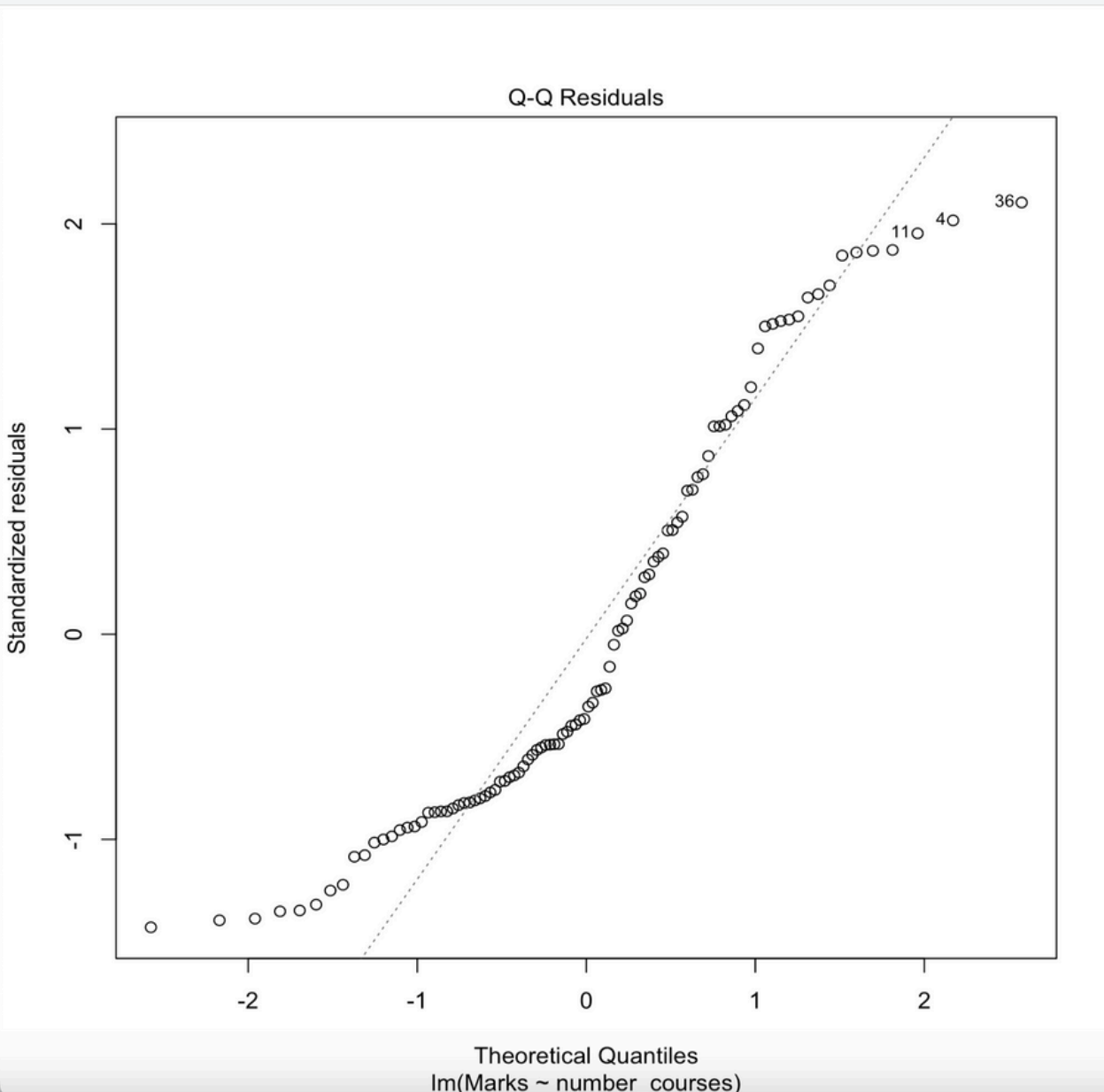
1. Linearly
2. Independence
3. Constant Variance
4. Normally Distributed Residuals

Limitations - Assess Whether the Model Assumptions are Met ?



As we can observe from the QQ plot and residual plot, the residual 1. have a constant variance, 2. is approximately normally distributed 3. is independent of one another. QQ plot shows good linearity
Thus we can conclude that the linear model we generate makes sense.

Limitations - Assess Whether the Model Assumptions are Met ?



Residual plot for regression of Marks on number_courses. As we can observe from the plot above, the plots show great linearity, independence, constant variance, and normally distributed residuals.